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	Critical to the success of long- term space flight is the prevention of human disease. Infectious diseases have impacted past missions. While the responsible microbes are carried onboard by the crew, it is not known if the response / adaptation of these microbes to this unique environment alters their pathogenic potential. Bacteria exhibit altered physiology and virulence properties in microgravity environments yet the mechanisms responsible for these adaptations remain unknown. In this study, we will investigate the effects of microgravity on the gene expression and virulence potential of the opportunistic pathogen S. pneumoniae using modeled microgravity and DNA microarray analysis. Objective 1: To identify and characterize differentially expressed S. pneumoniae genes cultured under modeled microgravity (MM) using total genomic DNA microarrays. Objective 2: To identify and characterize putative S. pneumoniae regulatory sequences of genes expressed in response to MM. Extended spaceflight or long-term habitation in space exposes humans to a number of environmental stresses such as ionizing radiation, microgravity, and prolonged confinement that can impact crew health and productivity. Alterations		

Task Description:	in immune function have been documented and include attenuations of the immune responses, which could increase crew susceptibility to infection (1-6). There is a documented history of infections during space flight that include those impacting the respiratory tract, skin, urinary tract, and gastrointestinal system (7). Bacteria responsible for such infections have often been associated with normal flora or carrier states, which express new properties or exploit a declining immune system as opportunistic pathogens (8-9). Evidence is accumulating that these bacteria undergo physiological and virulence-associated alterations in microgravity and the space environment (10-14). Changes observed in bacteria include increases in growth rate and biomass, structural changes in cell wall thickness, and reduced antibiotic sensitivity (10-13). Salmonella typhimurium, Proteus vulgaris, Bacillus subtilis, Escherichia coli, and Streptomyces levoris have displayed enhanced growth rates and biomass during spaceflight (10) while S. typhimurium and E. coli have exhibited an increased resistance to ionizing radiation (10-11). Further, Staphylococcus aureus and E. coli isolates have exhibited up to a two-fold reduction in antibiotic sensitivity compared to control isolates (10). The response of Salmonella to MM has also been recently investigated and an increase in some virulence properties reported (14). S. pneumoniae is a member of the normal flora in healthy individuals that colonizes the nasopharynx and can be isolated from 20-40% of children and 5-10% of adults. However, in immunocompromised individuals, it is a significant and important pathogen. The global rate of invasive pneumococcal disease is 15 in 100,000 individuals annually with the majority of cases found among newborns, the elderly, and other immunocompromised individuals. S. pneumoniae is a major cause of meningitis, bacterial pneumonia, bacteremia, otitis media, and sinusitis and is transmitted via aerosols. It has been isolated from crewmembers prior to space flight (11). Streptococcus species have also been isolated from air samples taken during shuttle flights and represent a potential health risk to crewmembers during extended space habitation (15). The virulence mechanism(s) of S. pneumoniae is being extensively investigated as well as the host defense mechanisms effective against these infections. However, little is known about S. pneumoniae and other opportunistic pathogens in terms of how they respond to microgravity and the other unique aspects of the space environment. In addition, little to nothing is understood as to the regulatory mechanisms behind bacterial adaptive responses to microgravity. Information on bacterial gene expression will advance our understanding of the regulation of bacterial virulence in this unique environment and the underlying pathogenic mechanisms. In this study, S. pneumoniae will be utilized to study the effects of microgravity on gene expression and begin to define regulatory mechanisms. This will be important to establish to generate a rational risk assessment of opportunistic bacterial disease in the space environment. References: 1. Manie, S., et al. 1991. Aviat. Space Env. Med. 62: 1153-1158. 2. Konstantinova, I. et al. 1993. J. Leukoc. Biol. 54: 189-201. 3. Taylor, G., Janney, R. 1992. J. Leukoc. Biol. 51: 129-132. 4. Taylor, G., Dardano, J. 1983. Aviat. Space Env. Med. 54[Suppl]: S55. 5. Talas, M. et al. 1983. Acta Microbiol. Hung. 30: 53. 6. Sonnenfeld, G. et al. 1990. Aviat. Space Env. Med. 61: 648. 7. Ferguson, J., Taylor, G., Mieszkuc, B. 1975. SP-368 Biomedical Results of Apollo. pp.83-103. 8. Nefedov, Y., Shilov, V., Konstantinova, I, Zalogoyev, S. 1971. Life Sci. Space Res. 9: 11-16. 9. Spizizen, J. 1971. Life Sci. Space Res. 9: 66-68. 10. Mishra, S., Pierson, D. 1992. Encyclopedia of Microbiology.Vol. 4. pp. 53-60. 11. Cioletti, L., Pierson, D., Mishra, S. 1991. SAE Technical Paper Series No.911512. 12. Belay, T. et al. 2002. J. Allergy Clin. Immunol. 110(2): 262-268. 13. Tixador R. et al. 1985. Acta Astronautica. 12: 131-134. 14. Nickerson, C.A. et al. 2000. Infect. Immun. 68 (6): 3147-3152. 15. Puleo, J.R., Oxborrow, G.S., Fields, N.D., Hall, H.E. 1970. Appl. Microbiol. 20 (3): 384-389.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from this study will increase our understanding of Streptococcus pneumoniae and its significance as an opportunistic human pathogen. These results will advance our understanding of the virulence mechanisms used by S. pneumoniae as it adapts to different environments. Understanding the responses of S. pneumoniae and other microbial pathogens in the space environment will aid in the development of risk assessments to address potential threats which could contribute to human disease during space flight. Information gained from this study will assist in understanding the threat of opportunistic infections which will be important in the planning and execution of long-term space missions and the habitation of space. Results from this study will contribute to the long-term goals of NASA's human space exploration directive and lead to the development of effective countermeasures crew members could use to detect and treat infections during space flight. In addition, direct earth benefits include gaining new knowledge about S. pneumoniae pathogenic mechanisms which could lead to the development of novel vaccines and therapeutics for this pathogen.
	Physiological and Virulence-based Responses to MM: Previous studies in our lab have investigated the physiological effects of MM on S. pneumoniae growth and chain length. Overnight cultures were diluted and grown in HARVs oriented in MM and 1 x g (gravity control) positions for 10 hours. Samples were collected at 100 minute intervals. Growth was assessed by monitoring optical density and quantitating colony forming units (CFUs) after serial dilution plating. Mean chain lengths were calculated by dividing total number of bacteria by total number of chains followed by statistical analysis using an unpaired Students t-test. No significant differences were observed in growth kinetics and mean chain lengths by bacteria grown under MM and 1 x g conditions. Environmental stress assays are currently underway to assess the effects of MM on resistance to thermal, acid, oxidative, and osmotic stresses. The effect of MM on S. pneumoniae virulence-potential was investigated by assessing changes in adherence and invasion in vitro and LD50 in mice. A549 lung epithelial monolayers were challenged with logarithmic-phase bacteria grown under MM and 1 x g. Adherence to and invasion in monolayers by MM-grown bacteria were found to be increased 4-fold and 1.4-fold, respectively. Interperitoneal (IP) challenges with MM and 1 x g cultured bacteria were carried out on 2 groups of 40 mice. Both groups were synchronously challenged and observed for 7 days. Results displayed a 3.5-fold decrease in LD50 (4.4 cfu) by bacteria grown under MM in comparison to 1 x g (15.2 cfu). Increases in virulence have previously been reported in murine challenges using MM-cultured Salmonella enterica serovar Typhimurium and extensions of these assays will be extrapolated in the investigation of an E. coli strain isolated from a Crohn’s Disease patient which has been recently classified in the new pathovar category of Adhesive-Invasive E. coli strains (AIEC) (1). S. pneumoniae Global Transcriptional Response to MM: MM-responsive genes constituting 14.4% of the S. pneumoniae TIGR4 genome were found to be differentially expressed = 1.5-fold based on microarray analysis at mid-logarithmic phases of growth. Ten percent of the responsive genes were found up-regulated by MM while 90% were down-regulated. The majority of MM-responsive genes were found to encode hypothetical proteins or transport and binding proteins. Most of the virulence-associated genes which were responsive to MM were down-regulated while only four were overexpressed (trpD, trpE, xseA, and SP0246). Two of the induced genes (trpD and trpE) are members of the Trp operon and encode key enzymes (anthranilate synthase and anthranilate phosphoribosyltransferase) involved in

Task Progress:	tryptophan biosynthesis, an essential amino acid used by microbes (2). The third upregulated gene, xseA, encodes the large subunit of the DNA repair enzyme, Exonuclease VII, which degrades single-stranded for the removal of mutations in genes (3). SP0246 encodes a homologue associated with the E. coli DeoR family of repressors which regulate the deoCABD operon involved in deoxyribonucleoside metabolism (4). Array results were found to correlate with those reported for Salmonella enterica serovar Typhimurium which contained MM-responsive genes located throughout the chromosome, most of which were respressed, but few known virulence genes responsive to MM (5). Differentially expressed groups of genes within operons and regulons included genes involved in natural DNA transformation (competence), systems involved in sugar uptake and transport (Phosphotransferase systems; PTS), and genes associated with the Bacteriocin-like peptide (Blp) regulon involved in the production of immunity and antimicrobial peptides (6-8). Data from this aim was recently presented at the 2004 American Society for Microbiology General Meeting held in New Orleans, LA and a manuscript is currently being prepared for submission which discusses these findings in further detail. Extensions of this aim to be conducted during the next year will include investigating effects of MM on global gene expression at other growth phases such as early (OD=0.2 to 0.3) and late (OD=0.7 to 1.0) logarithmic phases. Promoter Analysis of MM-responsive Genes: The Neural Network Promoter Predication Program (NNPP) was utilized to locate transcripton initiation sites of promoters within the upstream sequences of the MM-responsive genes (9). After these sites were identified, the Bacterial Promoter Recognition Program (BPROM) (Softberry, Inc.) was used to examine these regions for the presence of putative regulatory motifs. Binding motifs were identified which contained sequence homology with a number of known transcription factor binding sites (LyrR, NarL, ArgR, NagC, OmpR, LexA, MetJ, MetR, PhoB, CytR, DeoR) including binding motifs for several known global regulators (ArcA, Fnr, Fis, Crp, IHF). Interestingly, many of the motifs located within the promoters of MM-responsive genes included binding sites for members of the rpoD class of genes encoding sigma factors (s70) associated with heat shock stress responses (10). In a recent report rpoS (ss) concentrations were found to be two-fold higher in stationary cultures of E. coli grown under MM vs. 1 x g cultures. The MM-grown cultures were also found to be more resistant to general stresses such as hyperosmotic (2.5M NaCl) and acidic (pH 3.5) conditions (11). The sigma factors associated with identified motifs in this aim may prove to be critical regulators in increasing microbial resistance to various environment stresses under MM. Analyses proposed in this aim will be continued in the next year to further characterize these motifs and their associated transcriptional regulators. Goals will include the identification and characterization of conserved consensus sequences and novel uncharacterized binding motifs. This will be accomplished by investigating the location and frequency of identified motifs throughout the S. pneumoniae TIGR4chromosome and identifying the presence of novel motifs using alignment programs and searches within public databases containing known transcription factor binding sites. Transcriptional regulators associated with identified motifs will be investigated for known regulatory roles and to identify any interplay they may have in established regulatory networks to investigate potential roles in responses to MM. References: 1. C. A. Nickerson et al., Infect. Immun. 68, 3147 (June 1, 2000, 2000). 2. C. Kim et al., FEBS Letters 523, 239 (2002/07/17, 2002). 3. J. Chase, B. Rabin, J. Murphy, K. Stone, K. Williams, J. Biol. Chem. 261, 14929 (November 15, 1986, 1986). 4. P. Valentin-Hansen, Albrechtsen, B. , Love Larsen, J.E., EMBO J. 5, 2015 (August, 1986). 5. J. W. Wilson et al., PNAS 99, 13807 (October 15, 2002, 2002). 6. E. A. Campbell, S. Y. Choi, H. R. Masure, Mol Microbiol 27, 929 (March 01, 1998, 1998). 7. C. Siebold, K. Flukiger, R. Beutler, B. Erni, FEBS Letters 504, 104 (2001/8/31, 2001). 8. A. de Saizieu et al., J. Bacteriol. 182, 4696 (September 1, 2000, 2000). 9. M. G. Reese, Comput Chem. 26, 51 (December, 2001). 10. A. Kumar et al., Journal of Molecular Biology 235, 405 (1994/1/13, 1994). 11. S. V. Lynch, E. L. Brodie, A. Matin, J. Bacteriol. 186, 8207 (December 15, 2004, 2004).
Bibliography Type:	Description: (Last Updated: 06/28/2012)
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